

Task Scenario

The current scheduling scenario is Port. Port has 12 berths, 8 gantry cranes..., with their current statuses as follows:... . Current scheduling plan involves 12 vessels. Vessel 1 is scheduled to arrive at 8:00, requiring 2 hours for operations and 30 minutes of post-operation dwell time...



Function library



Function 1 : Master Scheduler
Purpose : Generate an overall scheduling plan...



Function 2 : Path Router
Purpose : Plan the optimal paths for entities...



Function 3 : Truck Allocator
Purpose : Assign trucks to transport containers ...



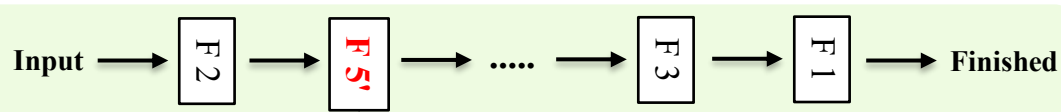
Function 4 : Crane Scheduler
Purpose : Allocate gantry cranes to vessels...



Function 5 : Yard Manager
Purpose : Distribute containers to yards...



Other Tool Functions
Purpose : Track real-time status of entities...



Normal Scheduling

Scheduling System



The code for the current functions is F1, F2, F3... Their respective functions are... As a senior architect, your task is to **refactor the code to achieve full decoupling**, ensuring it aligns with the current scenario...

Emergency Situations



Vessel arrival delay



Gantry crane breakdown



Forklift breakdown



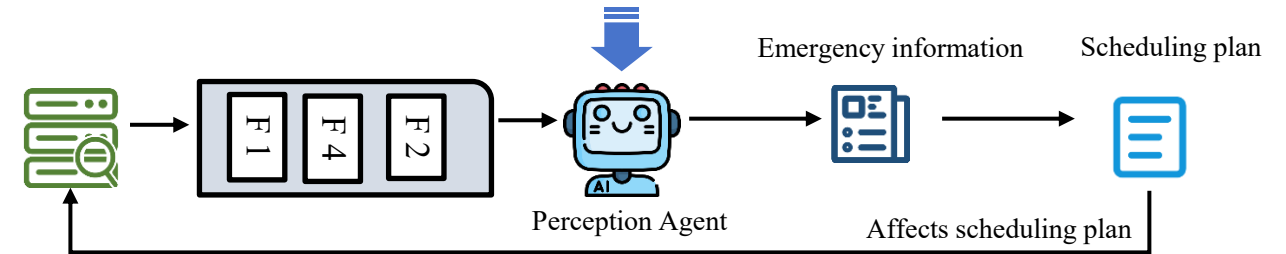
Internal port road congestion



Partial yard area unavailability

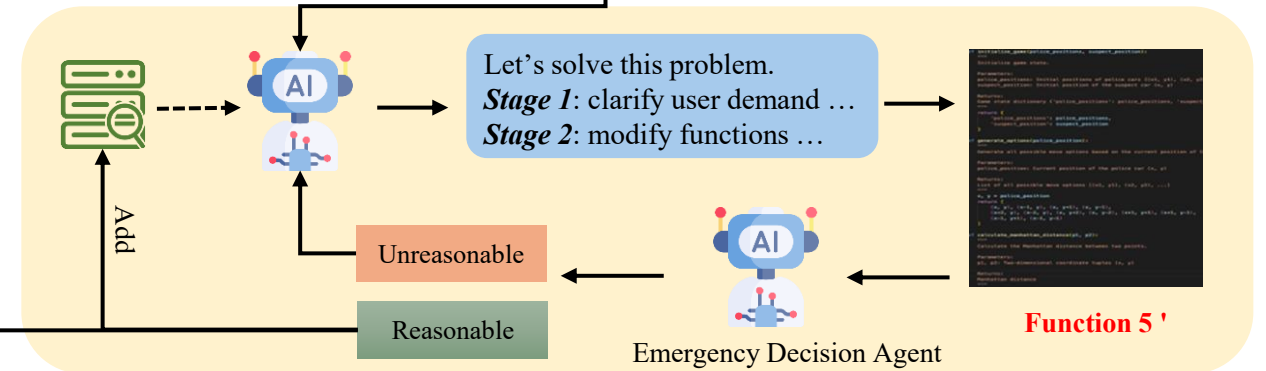


Yard crane malfunction



Function 5 Code

Emergency Information



Scheduling With Emergency situation